

Android Project Report for "Tour Guide"

1 Introduction

This report outlines the recent changes made to the Android application. The updates include modifications to the build configuration, changes in Java files, XML layout updates, and the introduction of new files.

2 Code Changes

2.1 build.gradle.kts

The `build.gradle.kts` file has been updated to use the Android Gradle plugin version 8.1.4. New dependencies have been added.

```
plugins {
    kotlin("android")
    kotlin("android.extensions")
}

android {
    ...
    buildFeatures {
        viewBinding = true
    }

    compileSdkVersion 30
    defaultConfig {
        ...
        minSdkVersion(16)
        targetSdkVersion(30)
    }

    ...

    dependencies {
        ...
        implementation("com.google.android.gms:play-services-maps:18.2.0")
        implementation("com.google.android.gms:play-services-location:21.0.1")
        implementation("com.squareup.retrofit2:retrofit:2.9.0")
        ... (other dependencies)
    }
}
```

2.2 ExploreFragment.java (Previously ActivityOne.java)

The file has been renamed from `ActivityOne.java` to `ExploreFragment.java`. The class has been updated to extend `Fragment` instead of `AppCompatActivity`. The `onCreate` method has been replaced with `onCreateView` for a `Fragment`. Context is now obtained using `requireContext()`. The `onDestroy()` method is now public and overridden for the `Fragment` lifecycle. Permission requests are made using `requireActivity()` for the fragment. The layout is inflated using `inflater.inflate(R.layout.activity_one,`

container, false). Markers are added using `MyMap.addMarker(markerOptions)`. Toast messages are displayed using `Toast.makeText(requireContext(), ...)`. Location Manager and Places are initialized using `requireContext()` and `createClient(requireContext())` for the fragment. Added marker click handling logic to initiate directions and draw polylines. Introduced the `OnDirectionsResultListener` interface for handling the results of directions requests. Updated the `getDirections`, `drawPolyline`, and `drawPolyline1` methods to incorporate directions functionality. Modified the `onMapReady` method to set a click listener for the map. Implemented the logic for saving a place to the database. Updated the `getLocation` method to handle location permissions and settings. Enhanced the `requestNewLocationData` method to update the current location.

```
package com.example.tourguide;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Toast;

import androidx.fragment.app.Fragment;

public class ExploreFragment extends Fragment {

    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container,
                             Bundle savedInstanceState) {
        // Inflate the layout for this fragment
        View view = inflater.inflate(R.layout.activity_one, container, false);

        // ... (Other modifications in ExploreFragment.java)

        return view;
    }

    // ... (Other methods and changes in ExploreFragment.java)
}
```

2.3 FavoritesFragment.java (Previously ActivityFour.java)

The file has been renamed from `ActivityFour.java` to `FavoritesFragment.java`. The class has been updated to extend `Fragment` instead of `AppCompatActivity`. The `onCreate` method has been replaced with `onCreateView` for a `Fragment`. Context is now obtained using `requireContext()`. The layout is inflated using `inflater.inflate(R.layout.activity_four, container, false)`. The `RecyclerView` is now accessed through the inflated view using `findViewById`. The `apiCalls` method now uses `requireContext()` when creating the `Volley RequestQueue`. The `mRecyclerView` and `layoutManager` are now initialized using `requireContext()`.

2.4 RestaurantsFragment.java (Previously ActivityTwo.java)

The file has been renamed from `ActivityTwo.java` to `RestaurantsFragment.java`. The class has been updated to extend `Fragment` instead of `AppCompatActivity`. The `onCreate` method has been replaced with `onCreateView` for a `Fragment`. Context is now obtained using `requireContext()`. The layout is inflated using `inflater.inflate(R.layout.activity_two, container, false)`. The `RecyclerView` is accessed through the inflated view using `inflate.findViewById(R.id.recyclerview)`. The `recyclerView.setLayoutManager` is now initialized with new `LinearLayoutManager(requireContext())`. The adapter for the `recyclerView` is set using new `MyAdapter(requireContext(), items)`.

2.5 SightsFragment.java (Previously ActivityThree.java)

The file has been renamed from `ActivityThree.java` to `SightsFragment.java`. The class has been updated to extend `Fragment` instead of `AppCompatActivity`. The `onCreate` method has been replaced with `onCreateView` for a `Fragment`. Context is now obtained using `requireContext()`. The layout is inflated using `inflater.inflate(R.layout.activity_three, container, false)`. The `RecyclerView` is accessed through the inflated view using `inflate.findViewById(R.id.recyclerview)`. The `recyclerView.setLayoutManager` is now initialized with new `LinearLayoutManager(requireContext())`. The adapter for the `recyclerView` is set using new `MyAdapter(requireContext(), items)`.

2.6 MainActivity.java

The `MainActivity.java` file has been modified. The class now includes fragment declarations for `ExploreFragment`, `RestaurantsFragment`, `SightsFragment`, and `FavoritesFragment`. The `onCreate` method has been updated to include fragment initialization and handling for `BottomNavigationView`. The bottom navigation item selection is now handled using a switch case for fragment loading. The buttons now launch corresponding fragments (`ExploreFragment`, `RestaurantsFragment`, `SightsFragment`, `FavoritesFragment`) instead of activities. A helper method `loadFragment(Fragment fragment)` has been added to streamline fragment loading.

```
package com.example.tourguide;

import android.os.Bundle;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentTransaction;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Fragment declarations
        ExploreFragment exploreFragment = new ExploreFragment();
        RestaurantsFragment restaurantsFragment = new RestaurantsFragment();
        SightsFragment sightsFragment = new SightsFragment();
        FavoritesFragment favoritesFragment = new FavoritesFragment();

        // Set default fragment
        loadFragment(exploreFragment);

        // BottomNavigationView handling
        BottomNavigationView bottomNavigationView = findViewById(R.id.bottom_navigation);
        bottomNavigationView.setOnNavigationItemSelectedListener(item -> {
            Fragment selectedFragment;

            // Switch case for fragment loading
            switch (item.getItemId()) {
                case R.id.navigation_explore:
                    selectedFragment = exploreFragment;
                    break;
                case R.id.navigation_restaurants:
                    selectedFragment = restaurantsFragment;
                    break;
                case R.id.navigation_sights:
                    selectedFragment = sightsFragment;
                    break;
                case R.id.navigation_favorites:
```

```

        selectedFragment = favoritesFragment;
        break;
    default:
        selectedFragment = exploreFragment;
    }

    // Load selected fragment
    loadFragment(selectedFragment);
    return true;
});
}

// Helper method to streamline fragment loading
private void loadFragment(Fragment fragment) {
    FragmentTransaction transaction = getSupportFragmentManager().beginTransaction();
    transaction.replace(R.id.fragment_container, fragment);
    transaction.addToBackStack(null);
    transaction.commit();
}

// ... (Other methods in MainActivity.java)
}

```

3 XML Layout Changes

3.1 activity_four.xml

The file has been updated to reference `FavoritesFragment` instead of `ActivityFour`.

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".FavoritesFragment">

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/my_recycler_view"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:padding="16dp"
        android:scrollbars="vertical"
        android:layout_above="@+id/bottom_navigation"
        android:layout_alignParentTop="true"
        android:layout_alignParentStart="true"/>

    <!-- Other layout elements or views can be added as needed -->

    <com.google.android.material.bottomnavigation.BottomNavigationView
        android:id="@+id/bottom_navigation"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_alignParentBottom="true"
        app:menu="@menu/bottom_navigation_menu"/>
</RelativeLayout>

```

3.2 activity_main.xml

New elements have been added, including a `FrameLayout` for fragment transactions and a `BottomNavigationView`.

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <!-- Other existing layout elements or views can be found here -->

    <ImageView
        android:id="@+id/logoImageView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:layout_marginBottom="30dp"
        app:layout_constraintBottom_toTopOf="@+id/bottom_navigation"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        android:visibility="gone"/>

    <FrameLayout
        android:id="@+id/fragment_container"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintBottom_toTopOf="@+id/bottom_navigation"/>

    <com.google.android.material.bottomnavigation.BottomNavigationView
        android:id="@+id/bottom_navigation"
        android:layout_width="0dp"
        android:background="@color/white"
        app:labelVisibilityMode="labeled"
        android:layout_height="wrap_content"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:menu="@menu/bottom_navigation_menu"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

3.3 activity_one.xml, activity_two.xml, activity_three.xml

Similar changes have been made to `activity_one.xml`, `activity_two.xml`, and `activity_three.xml`.

... (Details of changes in `activity_one.xml`, `activity_two.xml`, and `activity_three.xml`)

3.4 bottom_navigation_menu.xml

A new XML file for the bottom navigation menu has been created.

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
```

```

        android:id="@+id/navigation_explore"
        android:title="Explore"
        android:icon="@drawable/ic_explore" />

<item
    android:id="@+id/navigation_restaurants"
    android:title="Restaurants"
    android:icon="@drawable/ic_restaurants" />

<item
    android:id="@+id/navigation_sights"
    android:title="Sights"
    android:icon="@drawable/ic_sights" />

<item
    android:id="@+id/navigation_favorites"
    android:title="Favorites"
    android:icon="@drawable/ic_favorites" />

</menu>

```

3.5 themes.xml

The colorPrimary attribute in themes.xml has been modified.

```

<?xml version="1.0" encoding="utf-8"?>
<resources>
    <!-- Base application theme. -->
    <style name="Base.Theme.TourGuide" parent="Theme.AppCompat.Light.NoActionBar">
        <!-- Customize your light theme here. -->
        <item name="colorPrimary">@color/pale_olive</item>
    </style>

    <style name="Theme.TourGuide" parent="Base.Theme.TourGuide" />
</resources>

```

4 New Files

4.1 DirectionsApiService.java

A new file DirectionsApiService.java has been added.

```

package com.example.tourguide;

import retrofit2.Call;
import retrofit2.http.GET;
import retrofit2.http.Query;

public interface DirectionsApiService {
    @GET("maps/api/directions/json")
    Call<DirectionsResponse> getDirections(
        @Query("origin") String origin,
        @Query("destination") String destination,
        @Query("key") String apiKey
    );
}

```

4.2 DirectionsResponse.java

A new file `DirectionsResponse.java` has been added.

```
package com.example.tourguide;

import java.util.List;

public class DirectionsResponse {
    public List<Route> routes;
}
```

5 Modified Files

5.1 OnDirectionsResultListener.java, Polyline.java, Route.java

Changes have been made to these existing files to introduce new functionalities.

... (Details of changes in `OnDirectionsResultListener.java`, `Polyline.java`,
and `Route.java`)